

Unit 8 - Proving Relationships and Theorems	
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The following B.E.S.T Standards will be covered in this unit:

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships and theorems of parallelograms.

- *Clarification 1: Postulates, relationships and theorems include opposite sides are congruent, consecutive angles are supplementary, opposite angles are congruent, the diagonals of a parallelogram bisect each other, and rectangles are parallelograms with congruent diagonals.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships and theorems of trapezoids.

- *Clarification 1: Postulates, relationships and theorems include the Trapezoid Midsegment Theorem and for isosceles trapezoids: base angles are congruent, opposite angles are supplementary and diagonals are congruent.*
- *Clarification 2: Instruction includes constructing two-column proofs, pictorial proofs, paragraph and narrative proofs, flow chart proofs or informal proofs.*
- *Clarification 3: Instruction focuses on helping a student choose a method they can use reliably.*

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.

- *Clarification 1: Within the Geometry course, instruction focuses on the connection to proofs within the course.*

Unit 8, Lesson 1: Proving Properties of Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can define a parallelogram.
- ☐ I can prove opposite sides and opposite angles of a parallelogram are congruent.
- ☐ I can prove consecutive angles of a parallelogram are supplementary.
- ☐ I can prove diagonals of a parallelogram bisect each other.



8.1.1 Warm-Up: Cartographer

Dave Imus is a cartographer whose maps have won many awards. Dave sees the world through understanding places. Dave's maps are considered art.

1. Have you ever used a paper map?

2. Since you can't put everything on a map, what three places do you think should be on the map of Florida?



8.1.2 Collaboration: Parallel Lines and Four-Sided Figures

1. On a piece of patty paper, draw one set of parallel lines. Overlap your patty paper with a partner's parallel lines to create an enclosed figure. Each partner should trace the enclosed shape onto a new piece of patty paper.
 - a. Classify the enclosed shape.
 - b. Justify your classification.
 - c. Ask two classmates how they classified the figure they created. Record their classification and write your summary of their justification. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Classification	My Summary of Their Justification	Initials



8.1.3 Exploration: Properties of Parallelograms



A quadrilateral containing two pairs of parallel sides is a **parallelogram**. A member of the following shape classes: polygons, quadrilaterals, trapezoids. Sub-classes include rectangles, rhombi, and squares.

1. Using a ruler, measure the pairs of opposite sides in each parallelogram you and your partner created.
 - a. What do you notice?
 - b. Make a conjecture about the measures of opposite sides of a parallelogram.
2. Using a ruler, draw the diagonals of one of the parallelograms.
 - a. Determine the lengths of each diagonal.
 - b. Mark the point where the two diagonals intersect.

c. What do you notice about the point of intersection and its relationship to each diagonal?

d. Repeat the same steps to draw and measure the diagonals in your second parallelogram.

e. Make a conjecture about the diagonals of a parallelogram.

3. Using a protractor, measure the pairs of opposite interior angles of your parallelograms.

a. What do you notice?

b. Make a conjecture about the measures of opposite angles of a parallelogram.

c. Make a conjecture about the measures of consecutive angles of a parallelogram.

4. Based on your conjectures, select all of the statements about a parallelogram that are true.

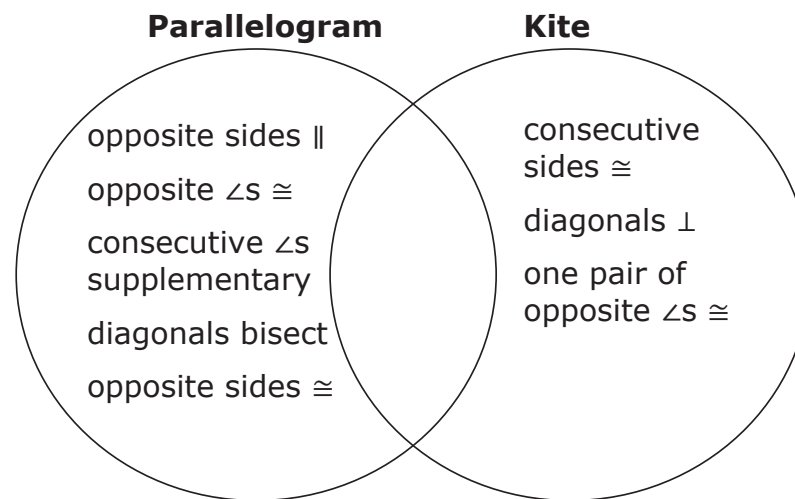
- ☐ A parallelogram has only one set of opposite parallel sides.
- ☐ A parallelogram has two sets of opposite parallel sides.
- ☐ A parallelogram has only one set of opposite congruent sides.
- ☐ A parallelogram has two sets of opposite congruent sides.
- ☐ Consecutive angles of a parallelogram are congruent.
- ☐ Consecutive angles of a parallelogram are supplementary.
- ☐ Opposite angles of a parallelogram are congruent.
- ☐ Opposite angles of a parallelogram are supplementary.
- ☐ Diagonals of a parallelogram bisect each other.



5. For each true statement in the previous problem, draw and name a parallelogram. Using the information in the statement, mark the diagram and write a mathematical statement. The first row has been completed.

Statement	Drawing	Given
A parallelogram has two sets of opposite parallel sides.		$\overline{AD} \parallel \overline{BC}$ $\overline{AB} \parallel \overline{DC}$

6. The Venn diagram shows the properties of a parallelogram and another quadrilateral called a kite. Discuss with your partner why a kite is not considered a parallelogram.



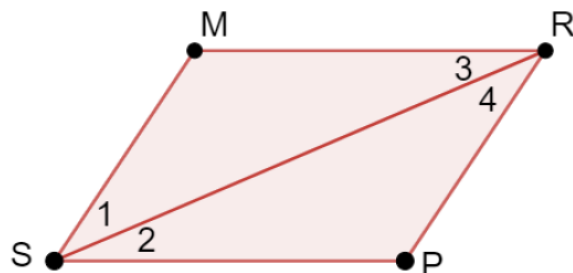
Theorems about Parallelograms

- Opposite sides of a parallelogram are congruent.
- Opposite angles of a parallelogram are congruent.
- Consecutive interior angles of a parallelogram are supplementary.
- The diagonals of a parallelogram bisect each other.



8.1.4 Guided Instruction: Proofs With Parallelograms

1. $MRPS$ is shown.



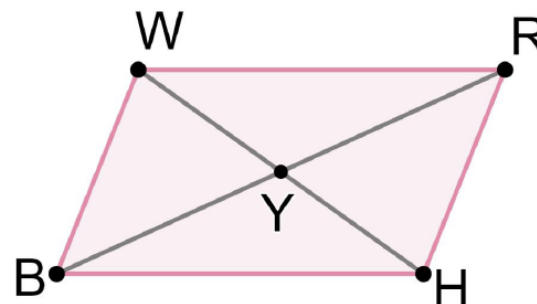
Given: $MRPS$ is a parallelogram

Prove: $\overline{MR} \cong \overline{PS}$ and $\overline{MS} \cong \overline{PR}$

Complete the following two-column proof.

Statement	Reason
1.	1. Given
2.	2.
3. $\angle 1 \cong \angle 4, \angle 2 \cong \angle 3$	3.
4. $\overline{SR} \cong \overline{SR}$	4. Reflexive property of congruence
5. $\triangle MSR \cong \triangle PRS$	5. ASA congruence theorem
6.	6. CPCTC

2. $WRHB$ with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y is shown.



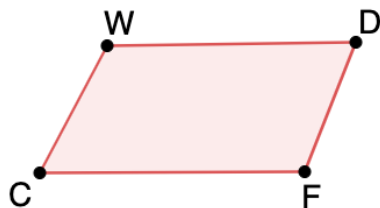
Given: $WRHB$ is a parallelogram with diagonals $\overline{WH}$ and $\overline{BR}$ intersecting at Y

Prove: $\overline{WH}$ and $\overline{BR}$ bisect each other.

Complete the following two-column proof.

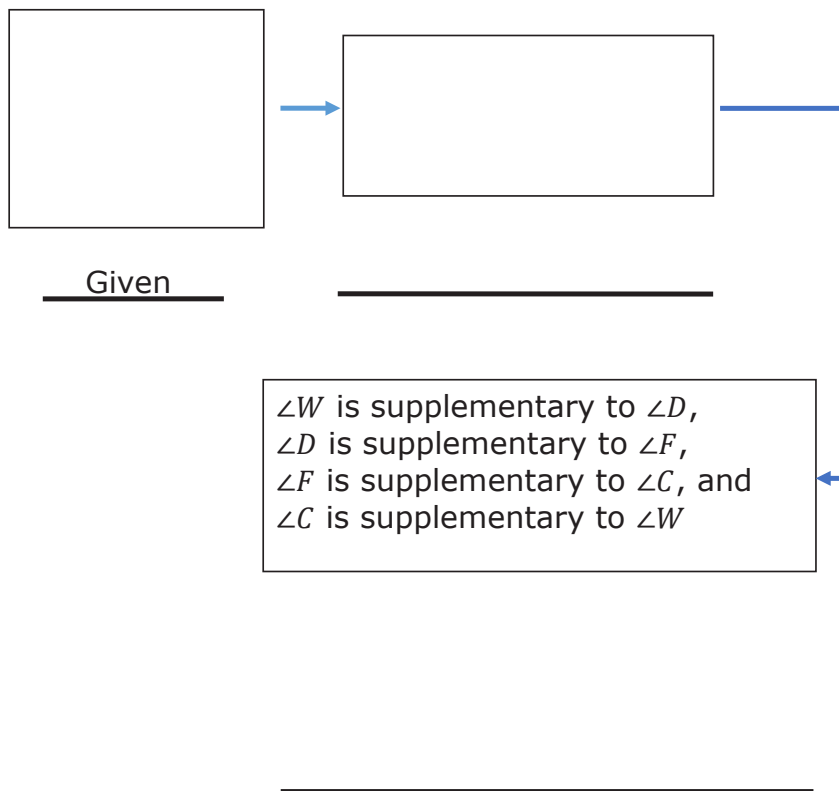
Statement	Reason
1.	1. Given
2.	2.
3. $\overline{RH} \cong \overline{BW}; \overline{WR} \cong \overline{HB}$	3.
4. $\angle WRB \cong \angle HBR;$ $\angle BHW \cong \angle RWH$	4.
5. $\triangle BYH \cong \triangle RYW$	5. ASA congruence theorem
6.	6. CPCTC
7. $BY = YR$ and $WY = YH$	7.
8.	8. Definition of a segment bisector

3. Complete the following flowchart proof. Statements go in the boxes and reasons go below the boxes.



Given: $WDFC$ is a parallelogram.

Prove: $\angle W$ is supplementary to $\angle D$,
 $\angle D$ is supplementary to $\angle F$,
 $\angle F$ is supplementary to $\angle C$, and
 $\angle C$ is supplementary to $\angle W$



8.1.5 Wrap-Up: Cool Down

- Write down one fact you knew about parallelograms before the lesson.
- Write down one fact you learned about parallelograms during the lesson.

Unit 8, Lesson 2: Using Properties of Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Target

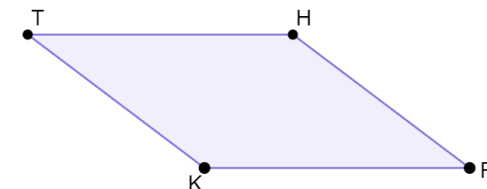
- ☐ I can determine if a quadrilateral is a parallelogram using properties of a parallelogram.



8.2.1 Warm-Up: Bell Work

Given: $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$

Prove: $THRK$ is a parallelogram



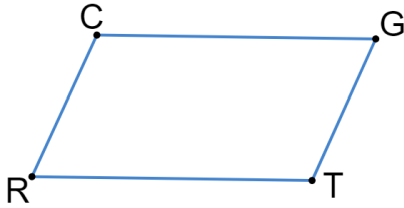
Statement	Reason
1. $\angle RKT \cong \angle RHT$ and $\angle KTH \cong \angle KRH$	1. Given
2. $m\angle RKT = m\angle RHT$ and $m\angle KTH = m\angle KRH$	2. Definition of Congruence
3.	3. The sum of the angles in a quadrilateral is 360°
4. $m\angle RKT + m\angle RKT + m\angle KTH + m\angle KTH = 360^\circ$ $m\angle RHT + m\angle RHT + m\angle KRH + m\angle KRH = 360^\circ$	4.
5. $2m\angle RKT + 2m\angle KTH = 360^\circ$ $2m\angle RHT + 2m\angle KRH = 360^\circ$	5. Addition
6.	6.
7. $\angle RKT$ is supplementary to $\angle KTH$. $\angle RHT$ is supplementary to $\angle KRH$	7.
8.	8. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
9. $THRK$ is a parallelogram	9.



8.2.2 Collaboration: Determining Parallelograms

Work with your partner to determine if the figure given is a parallelogram. Finish the statements that follow.

1. $CGTR$ is shown. In the figure, $m\angle C = 113^\circ$, $m\angle G = 67^\circ$ and $m\angle T = 103^\circ$.



a. I can conclude ...

b. I know this because ...

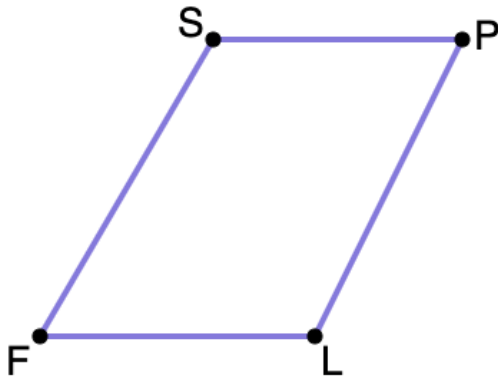
2. $WCXN$ is shown with sides $\overline{WC}$ and $\overline{NX}$ each measuring 15 cm and sides $\overline{WN}$ and $\overline{CX}$ each measuring 9 cm.



a. I can conclude ...

b. I know this because ...

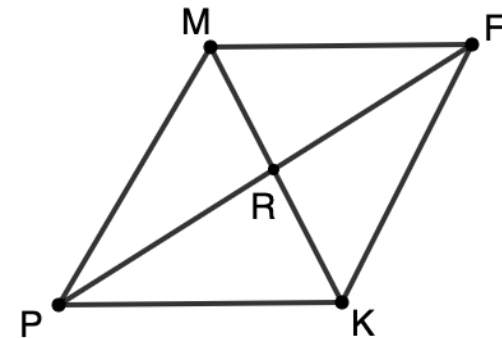
3. In $SPLF$, $m\angle S = 120^\circ$, and $m\angle F = (20x + 47)^\circ$. If $x = \frac{1}{2}$, determine whether the figure is a parallelogram.



a. I can conclude ...

b. I know this because ...

4. $MFKP$ with diagonals $\overline{MK}$ and $\overline{PF}$ intersecting at point R is shown.



Given that $MR = 4x + 7$, $KR = 3y - 2$, $FR = 4y - 3$, and $PR = 7x - 1$, find the values of x and y that make $MFKP$ a parallelogram.

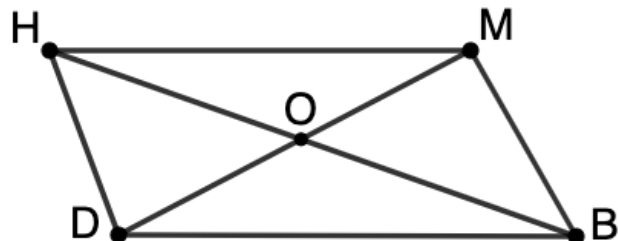
$$x = \underline{\hspace{2cm}} \quad y = \underline{\hspace{2cm}}$$

5. When determining if a figure is a parallelogram, what qualities must it have? List them below.



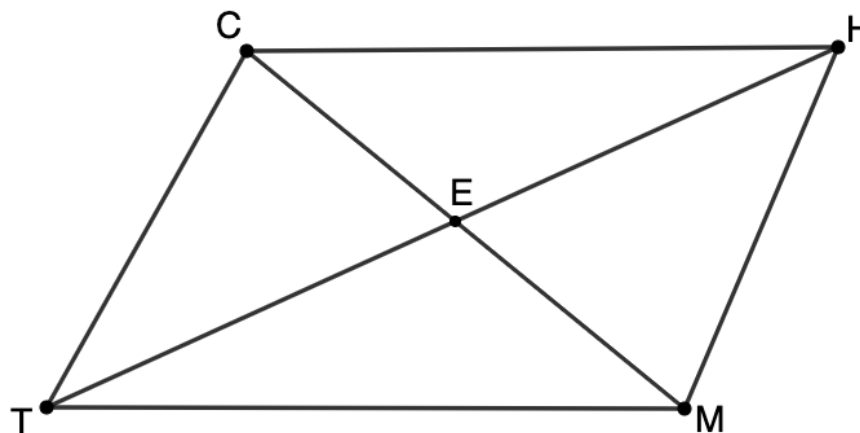
8.2.3 Guided Practice: Is It a Parallelogram?

1. Use the figure given to fill in the chart below.



If you knew ...	...would $HMBD$ be a parallelogram?	Why or why not?
$\angle DHO \cong \angle MBO$ and $\angle BDO \cong \angle HMO$	☐ yes ☐ no	
$\overline{HM} \cong \overline{DB}$	☐ yes ☐ no	
O is the midpoint of $\overline{DM}$ and $\overline{HB}$	☐ yes ☐ no	
$\angle DHO + \angle BDO = 180^\circ$	☐ yes ☐ no	

2. Rocio recently inherited a large field in the shape of a quadrilateral from her Uncle Asher. $CHMT$ with diagonals $\overline{CM}$ and $\overline{HT}$ intersecting at point E is shown. She wants to partition the field into four triangular sections to make gardens. While she was out measuring for her supply order, she noticed that the field was shaped like a parallelogram. Her Uncle Asher disagrees. Rocio created the diagram below with the following measurements: $m\angle CHT = 23^\circ$, $m\angle CET = 72^\circ$, $m\angle CMT = 49^\circ$, and $m\angle HTM = 23^\circ$.

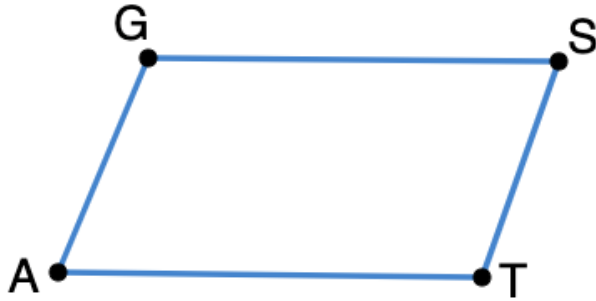


- a. Who is correct?
- b. How do you know?



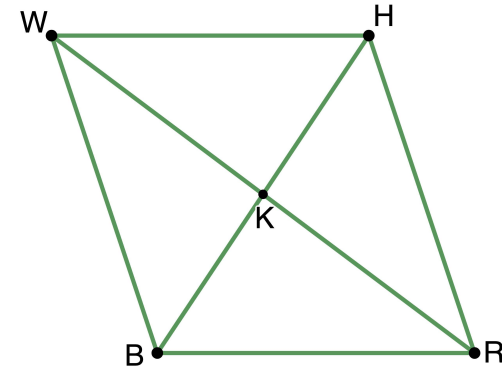
8.2.4 Try It: Parallelograms and Algebra

1. Quadrilateral $GSTA$ is shown where $GS = 3x - 1$, $ST = 2y - 3$, $AT = 11$, and $GA = 8$.



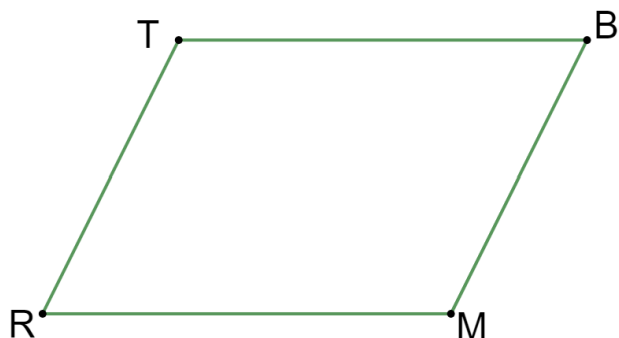
- a. If $x = 4$ and $y = 6$, determine whether $GSTA$ is a parallelogram.
- b. If $GSTA$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?

2. $WHRB$ is shown, where diagonals $\overline{WR}$ and $\overline{HB}$ intersect, such that $WK = -y + 4$, $RK = \frac{1}{3}x + 5$, $HK = 5x - 6$, and $BK = 11 + y$.



- a. If $x = 3$ and $y = -2$, determine whether $WHRB$ is a parallelogram.
- b. If $WHRB$ is not a parallelogram, what value(s) for x and y would make it a parallelogram?

3. Tariq and Agnes were having a discussion about the different properties of parallelograms. Together they drew and labeled $TBMR$, where $m\angle RTB = (5x - 2)^\circ$, $m\angle TBM = y^\circ$, and $m\angle BMR = 118^\circ$. What are the values of x and y for Tariq and Agnes' figure to be a parallelogram?



8.2.5 Wrap-Up: Cool Down

1. Check the property that is always true about the polygon.

Properties of Figure	Types of Polygons	
	Quadrilateral	Parallelogram
Only one set of parallel sides	☐	☐
Two sets of parallel sides	☐	☐
Two sides of equal length	☐	☐
Consecutive angles are supplementary.	☐	☐
Diagonals are congruent.	☐	☐
The sum of all angles is 360° .	☐	☐
Diagonals bisect each other.	☐	☐
Opposite angles are congruent.	☐	☐

Unit 8, Lesson 3: Special Parallelograms



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



Learning Targets

- ☐ I can prove rectangles are parallelograms.
- ☐ I can use properties of parallelograms to identify special parallelograms.



8.3.1 Warm-Up: Estimation

1. A picture of a bowl with a cereal square is shown.



- a. Estimate the number of cereal squares that will go around the inner circumference of the bowl.

2. A picture of the bowl with a diameter of cereal squares is shown.

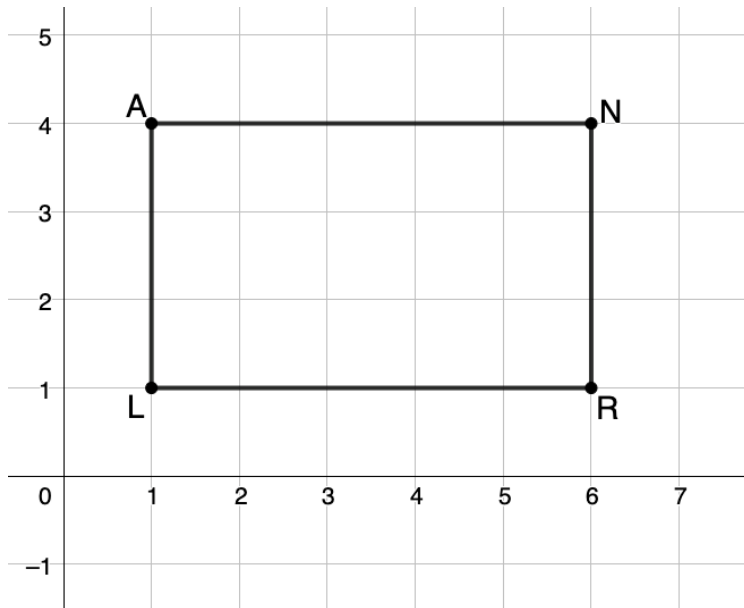


- a. Use the diameter of cereal squares to find the circumference.
 - b. Compare your estimate to the circumference found using the diameter of cereal squares.



8.3.2 Collaboration: What Is a Rectangle?

1. $ANRL$ is a rectangle shown on the coordinate grid.



- a. List everything you know about this type of quadrilateral.
- b. Compare your list with your partner's list. Add anything you may have missed.

- c. With your partner, list two truths and one lie about this type of quadrilateral in the table below. Share your list with another pair. See if they can spot the lie.

1.
2.
3.

2. $ZNHK$ is shown.



- a. Using a ruler, draw $\overline{ZH}$ and $\overline{KN}$.
- b. These segments are called _____.

- c. Measure the length of these segments and write them in the table below. Include units.

Segment	Length
$\overline{ZH}$	
$\overline{KN}$	

- d. Make a conjecture about the lengths of the diagonals of a rectangle.
- e. Label the intersection point of the diagonals S .
- f. Measure the following segments and fill in their length in the table below. Include units.

Segment	Length
$\overline{ZS}$	
$\overline{HS}$	
$\overline{KS}$	
$\overline{NS}$	

- g. What do you notice?
- h. Make a conjecture about what happens to the diagonals of a rectangle when they intersect.



8.3.3 Exploration: Comparing Parallelograms and Rectangles

1. Work with your partner to fill in the table. If the statement describes a property always true of the shape, mark the box.

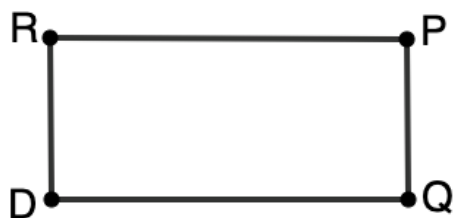
Statement	Parallelogram	Rectangle
opposite sides congruent	☐	☐
opposite sides parallel	☐	☐
opposite angles congruent	☐	☐
consecutive angles supplementary	☐	☐
diagonals bisect each other	☐	☐
diagonals are congruent	☐	☐
all angles are congruent	☐	☐

Based on the chart, discuss with your partner whether a rectangle could be classified as a parallelogram.



A **rectangle** is a quadrilateral that contains four right angles. Rectangles may be oblong or square. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $RPQD$ is a rectangle.



Complete the statements.

It is given that $RPQD$ is a rectangle. By the definition of

a rectangle $\angle R, \angle P, \angle Q$, and $\angle D$ are all

- ☐ right angles,
- ☐ acute angles,
- ☐ obtuse angles,

which means that $m\angle R = m\angle P = m\angle Q = m\angle D =$

- ☐ 90° .
- ☐ 180° .

Since $m\angle P + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

and since $m\angle D + m\angle Q = 180^\circ$ we can show that

- ☐ $\overline{RD} \parallel \overline{PQ}$
- ☐ $\overline{RP} \parallel \overline{DQ}$

by the

- ☐ alternate interior angles theorem.
- ☐ consecutive interior angles theorem.

Therefore, $RPQD$ is a

- ☐ parallelogram
- ☐ kite

by definition.



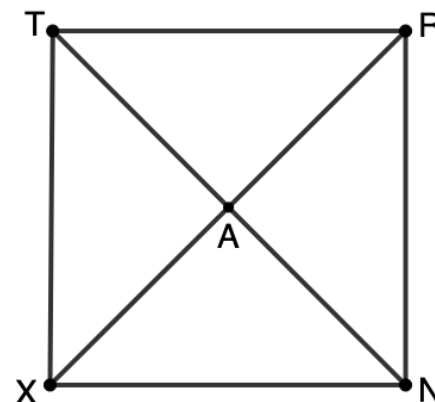
8.3.4 Guided Instruction: More Special Parallelograms



A **square** is a quadrilateral with four right angles and four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms, rectangles, rhombi.

1. $TRNX$ is a square with diagonals $\overline{TN}$ and $\overline{XR}$ intersecting at point A .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include the units.



Segment	Measure	Segment/Angle	Measure
$\overline{TN}$		$\overline{XA}$	
$\overline{XR}$		$\overline{RA}$	
$\overline{TA}$		$\angle XAT$	
$\overline{NA}$		$\angle NAR$	

- b. Which of the following describe the diagonals of a square? Circle all that apply.

congruent

bisect

perpendicular

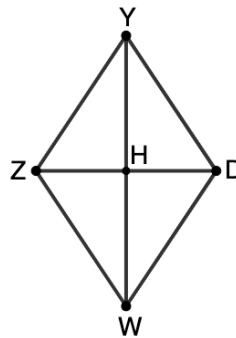
- c. List the features that determine if a parallelogram is a square.



A **rhombus** is a quadrilateral containing four equal-length sides. It's a member of the following shape classes: polygons, quadrilaterals, trapezoids, parallelograms. Squares form a sub-class.

2. $YDWZ$ is a rhombus with diagonals $\overline{YW}$ and $\overline{ZD}$ intersecting at point H .

- a. Use a ruler and a protractor to fill in the chart with the appropriate measurements. Include units.



Segment	Measure	Segment/Angle	Measure
$\overline{YW}$		$\overline{ZH}$	
$\overline{ZD}$		$\overline{DH}$	
$\overline{YH}$		$\angle ZHW$	
$\overline{WH}$		$\angle YHD$	

- b. Which of the following describe the diagonals of a rhombus? Select all that apply.

congruent

bisect

perpendicular

- c. List the features that determine if a parallelogram is a rhombus.

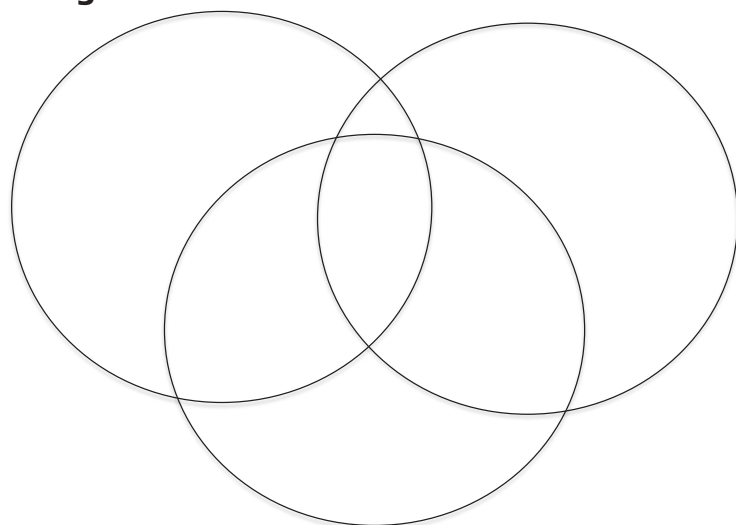
- d. What are the differences between a square and a rhombus?

3. Use the numbered statements in the box to fill in the Venn diagram comparing rectangles, squares, and rhombi.

- | | |
|----------------------------------|---|
| 1. opposite sides are congruent | 6. consecutive angles are supplementary |
| 2. opposite sides are parallel | 7. all sides are congruent |
| 3. all angles are congruent | 8. diagonals are congruent |
| 4. diagonals are perpendicular | 9. diagonals bisect each other |
| 5. opposite angles are congruent | |

Rectangle

Rhombus



Square



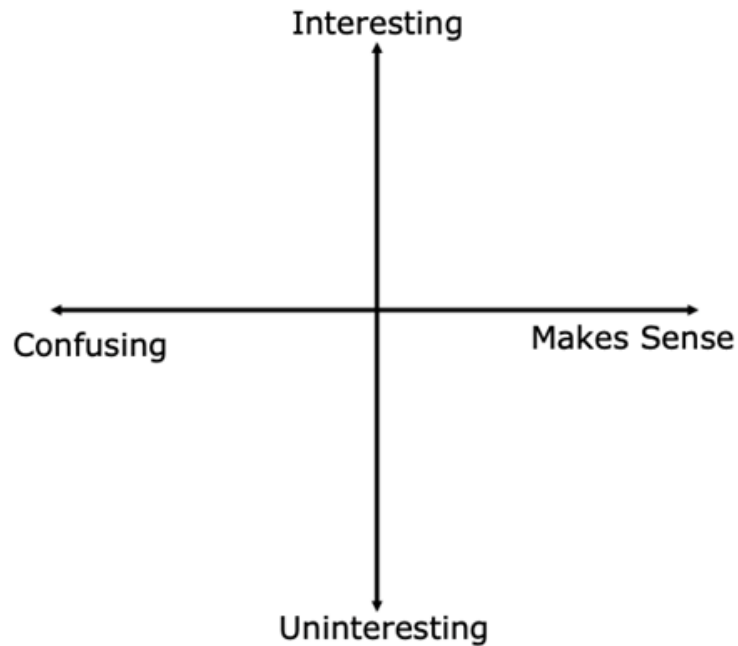
8.3.5 Bring It Together: Which Special Parallelogram Is It?

1. Explain how to show a parallelogram is a rectangle but not a rhombus or a square.
2. Explain how to show a parallelogram is a rhombus but not a square or a rectangle.
3. Explain how to show a parallelogram is a square.



8.3.6 Wrap-Up: Reflection

1. Plot a point on the grid to indicate your level of interest and how much sense you made of today's learning targets.



2. Explain why you placed the point where you did.

Unit 8, Lesson 4: Kites and Parallelograms



Benchmark

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.



Learning Target

- ☐ I can use properties of parallelograms, kites, and special parallelograms to solve mathematical and real-world problems.



8.4.1 Warm-Up: Canstruction

Each year just before Thanksgiving there are competitions held all over the world to build unique Canstructions with canned food that is collected from the community.



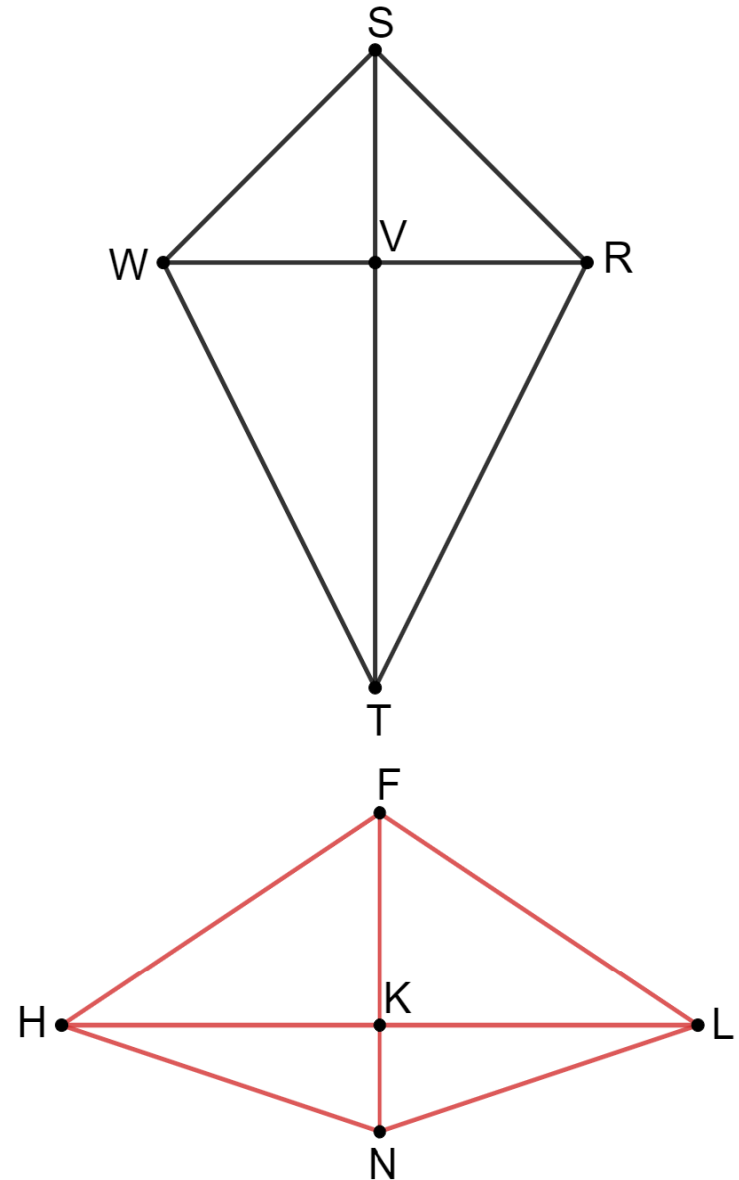
The food is then given to local food banks. The picture shows the Canstruction called "Downside Up," which was the winning exhibit from Canstruction 2010 in New York City.

1. What do you wonder about "Downside Up"?
2. Make an educated guess at the number of cans used to build "Downside Up". Show your work.
3. Compare your educated guess with your partner's. Describe how close your guess was to your partner's guess.



8.4.2 Exploration: What Is a Kite?

Kite $SRTW$ with diagonals $\overline{ST}$ and $\overline{WR}$ intersecting at point V and $FLNH$ with diagonals $\overline{FN}$ and $\overline{HL}$ intersecting at point K are shown.



- Using a ruler, work with a partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRTW</i>		<i>FLNH</i>	
Segment	Measurement	Segment	Measurement
$\overline{SR}$		$\overline{FL}$	
$\overline{SW}$		$\overline{FH}$	
$\overline{RT}$		$\overline{HN}$	
$\overline{WT}$		$\overline{LN}$	
$\overline{ST}$		$\overline{FN}$	
$\overline{SV}$		$\overline{FK}$	
$\overline{TV}$		$\overline{NK}$	
$\overline{WR}$		$\overline{HL}$	
$\overline{WV}$		$\overline{HK}$	
$\overline{RV}$		$\overline{LK}$	

- What do you notice about $\overline{SR}$ and $\overline{SW}$ in *SRTW* and $\overline{FL}$ and $\overline{FH}$ in *FLNH*?
- What do you notice about $\overline{RT}$ and $\overline{WT}$ in *SRTW* and $\overline{HN}$ and $\overline{LN}$ in *FLNH*?

- These segment pairs are **adjacent** sides of a kite.
 - Make a conjecture about the measures of adjacent sides of a kite.
 - Make a conjecture about the lengths of diagonals of a kite.
- Using a protractor, work with your partner to fill in the chart below with the appropriate measurements. Each partner should find the measurements for one kite and share their results with the other.

<i>SRXW</i>		<i>FLNH</i>	
Angle	Measurement	Angle	Measurement
$\angle SVR$		$\angle FKL$	
$\angle WSR$		$\angle HFL$	
$\angle RTW$		$\angle LNH$	
$\angle TWS$		$\angle FHN$	
$\angle TRS$		$\angle FLN$	

- Make a conjecture about what happens when the diagonals of a kite intersect.
- Make a conjecture about the measures of opposite angles of a kite.

A **kite** is a quadrilateral with exactly two pairs of adjacent congruent sides.



A **kite** has the following properties:

- One pair of opposite congruent angles
- Perpendicular diagonals
- One bisected diagonal



8.4.3 Collaboration: Quadrilaterals

- With your partner, use the bank of phrases to fill in the chart comparing quadrilaterals. Statements can be used more than once.

A. Opposite sides are congruent.	B. Opposite sides are parallel.
C. All sides are congruent.	D. Diagonals bisect each other.
E. Opposite angles are congruent.	F. Diagonals are congruent.
G. All angles are congruent.	H. Diagonals are perpendicular.
I. Two pairs of adjacent sides are congruent.	J. One diagonal is bisected.
K. One pair of opposite angles are congruent.	L. Opposite angles are supplementary.
M. One pair of parallel sides	N. Non-parallel sides are congruent.
O. Two pairs of congruent angles	

- Write the letter(s) from the bank.

Quadrilateral	Sides	Angles	Diagonals
Parallelogram			
Rectangle			
Square			
Rhombus			
Kite			

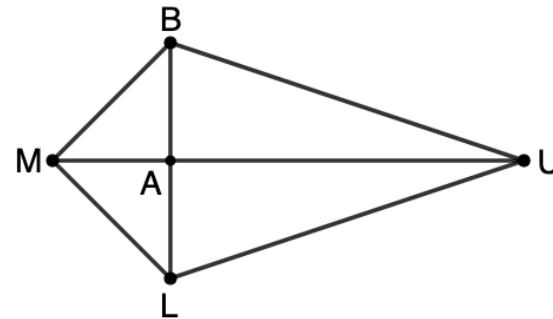
- b. Compare your chart with that of another pair. Add in anything you may have missed.

- c. Discuss with your partner whether or not kites can be classified as parallelograms.

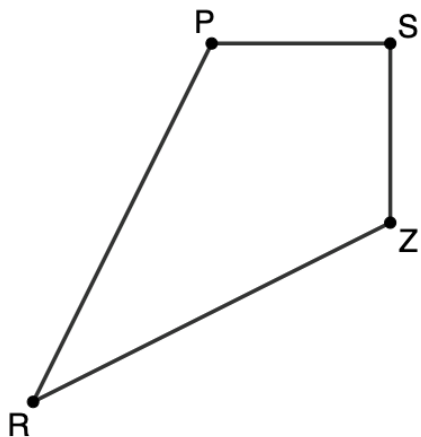


8.4.4 Guided Instruction: Quadrilaterals and Algebra

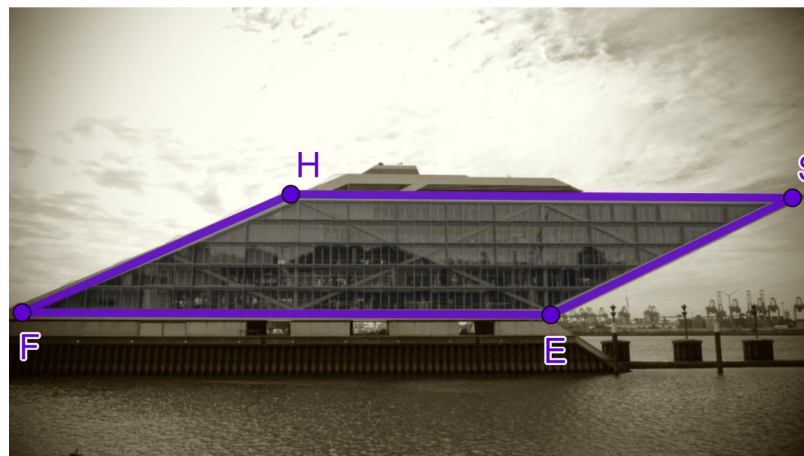
1. $MBUL$ with diagonals $\overline{BL}$ and $\overline{MU}$ intersecting at point A is shown. Find the value of x that makes $MBUL$ a kite if $BA = 3x - 1$, $LA = 17$, $MA = 2x$, and $UA = 23$.



2. Find the values of x and y that make $PSZR$ a kite if $PS = 5x - 4$, $PR = 8y - 4$, $SZ = 3y + 1$, and $RZ = 36$.



3. The Dockland office building is one of the most famous buildings in Hamburg, Germany. Built in 2005, this building houses roughly 9,000 square meters of office space and breathtaking views of the Elbe River, but the size and the views are not what makes this building so famous. The building itself is in the shape of a parallelogram, $HSEF$.

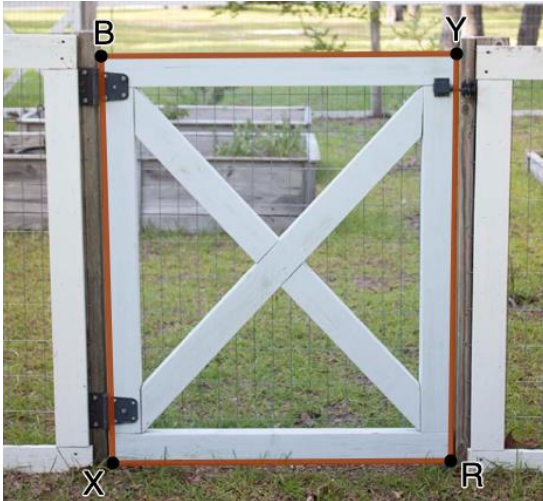


The perimeter of the Dockland building measures approximately 238 meters. If the length of $\overline{FE}$ is 2.8 times the length of $\overline{FH}$, what are the measures of $\overline{HS}$, $\overline{SE}$, $\overline{FE}$, and $\overline{FH}$? Round your answers to the nearest thousandth of a meter.



8.4.5 Your Turn!

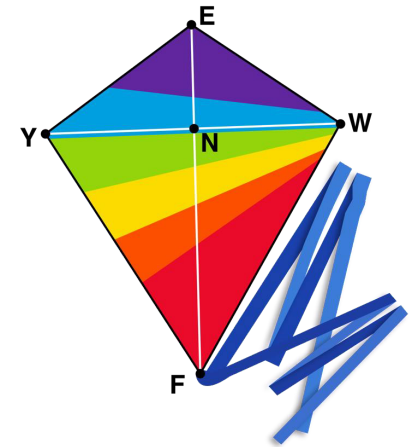
1. Oliver's carpentry company, Wood 4 You, has been hired to build a fence around a vegetable garden. The gate must be a perfect rectangle. The gate Oliver built is shown, where quadrilateral $BYRX$ outlines the gate.



Oliver wants to be absolutely sure that the gate is a rectangle. Describe two different ways Oliver can use his tape measure to ensure the gate is a true rectangle.

2. Piper is entering a kite-making contest. The kite she designed is shown, where the kite has been labeled as $YEW F$ with measurements $EW = 7.5$, $YN = 6$, and $FN = 10.2$, given in inches.

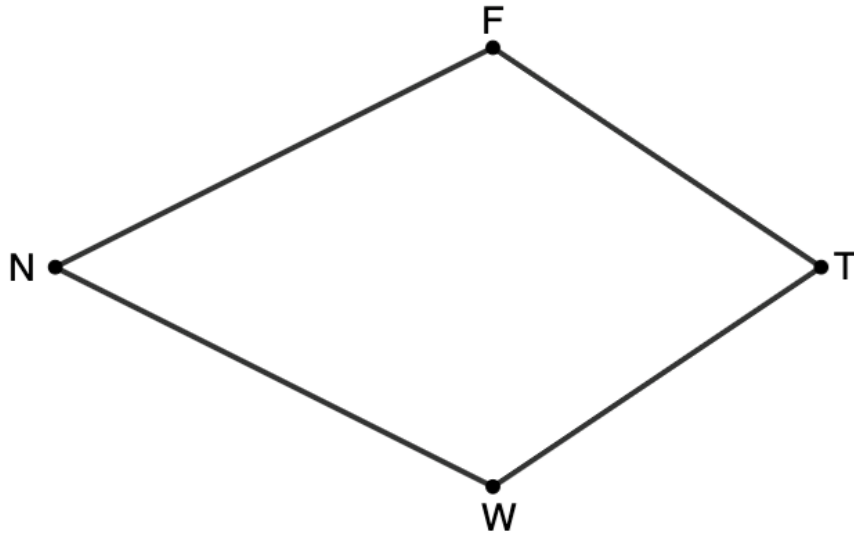
Piper wishes to add a strip of glitter fabric along diagonals $\overline{EF}$ and $\overline{YW}$ that will overlap at their intersection point N , as well as along the perimeter of the kite. How many inches of glitter fabric are needed to cover the diagonals and the perimeter of the kite?





8.4.6 Wrap-Up: Cool Down

1. Kite $FTWN$ is shown. Find the value of x if $m\angle N = 53^\circ$, $m\angle T = 67^\circ$, and $m\angle F = \left(\frac{1}{2}x - 8\right)^\circ$.



Unit 8, Lesson 5: Trapezoids



Benchmarks

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.

MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.



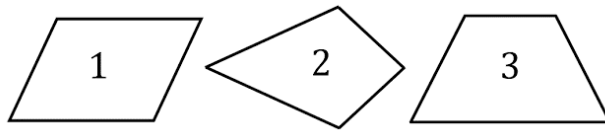
Learning Targets

- ☐ I can prove relationships and theorems about trapezoids.
- ☐ I can use relationships and theorems about trapezoids.



8.5.1 Warm-Up: Trapezoid or Trapezium?

1. Which quadrilateral is classified as a trapezoid?



There is more than one definition of a trapezoid.

Inclusive Definition	Exclusive Definition
A quadrilateral with at least one pair of parallel opposite sides.	A quadrilateral with exactly one pair of parallel opposite sides.

There is more than one definition of trapezium.

American Definition	British Definition
A quadrilateral with no parallel sides.	A quadrilateral with two sides parallel.

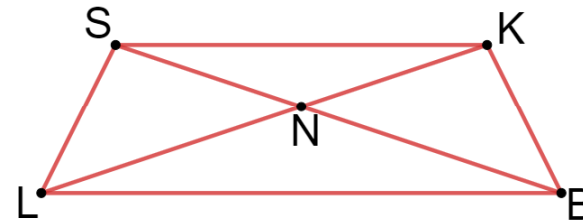
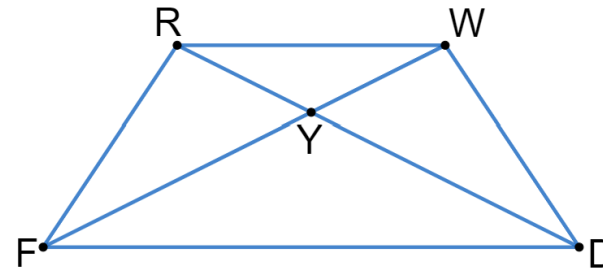
2. Use the definitions to classify quadrilaterals 1, 2, and 3. Write the numbers of the quadrilateral in the table.

Inclusive Definition of Trapezoid	Exclusive Definition of Trapezoid
American Definition of Trapezium	British Definition of Trapezium



8.5.2 Collaboration: Properties of Isosceles Trapezoids

1. $RWDF$ and $SKBL$ are both **isosceles trapezoids**. $\overline{RD}$ and $\overline{WF}$ are the diagonals of $RWDF$ and intersect at point Y . $\overline{SB}$ and $\overline{KL}$ are the diagonals of $SKBL$ and intersect at point N .



- a. Using a ruler and a protractor to measure $RWDF$ and $SKBL$, work with your partner to determine if the following conjectures about isosceles trapezoids are true or false. If false, write a new conjecture based on your measurements.

Conjecture	True or False?	If false, write a new conjecture.
All angles of an isosceles trapezoid are congruent		
Opposite sides are parallel		
Opposite angles are congruent		
Opposite sides are congruent		
Diagonals bisect each other		

- b. Compare your chart to that of another pair. Discuss any differences. Make corrections as necessary.

2. $\angle RFD$ and $\angle WDF$ in $RWDF$ are called **base angles**. $\angle SLB$ and $\angle KBL$ are base angles in $SKBL$.

Based on your previous measurements, make a conjecture about the base angles of an isosceles trapezoid.

3. Each isosceles trapezoid has two pairs of base angles.
- What is the other pair of base angles in trapezoid $RWDF$?
 - What is the other pair of base angles in trapezoid $SKBL$?



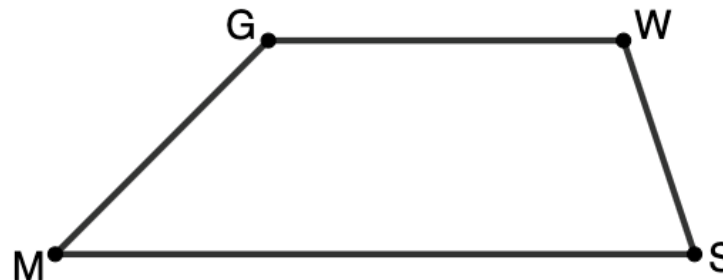
A **trapezoid** is a quadrilateral with *at least* one pair of parallel sides. It's a member of the following shape classes: polygons, quadrilaterals. Its sub-classes include parallelograms, rectangles, rhombuses, and squares.

An **isosceles trapezoid** is a trapezoid with at least one pair of congruent base angles.



8.5.3 Exploration: Parts of a Trapezoid

1. Trapezoid $GWSM$ is shown.



- a. The parallel sides of a trapezoid are called the **bases**. The trapezoid shown has one pair of bases.

Label each base red and mark them to show they are parallel on $GWSM$.

- b. The non-parallel sides of a trapezoid are called the **legs**. The trapezoid shown has two legs.

Label each leg of $GWSM$.



The **median** of a trapezoid connects the non-parallel sides at their midpoints. The median is sometimes referred to as the **midsegment**.

- c. Using your ruler, plot the midpoint of $\overline{GM}$ and the midpoint of $\overline{WS}$. Connect these points to create the median of trapezoid $GWSM$ in green. Be sure to mark the legs appropriately showing that the median is connected at the midpoints.

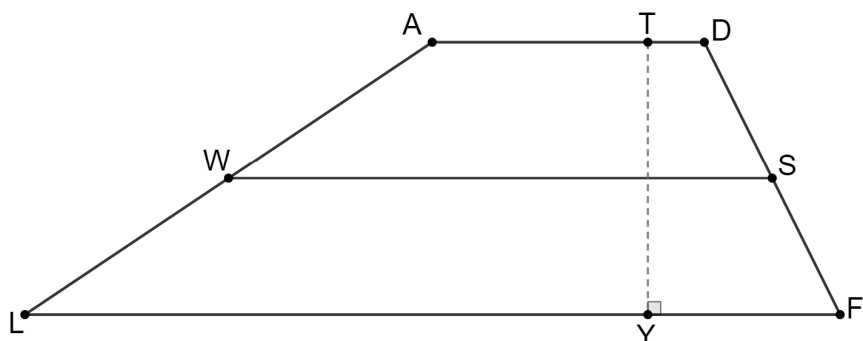


An **altitude** of a trapezoid is any segment from a point on one of the bases perpendicular to the line containing the other base.

- d. The length of the altitude is the distance between the bases.

Draw the altitude on $GWSM$. Name the altitude by naming the endpoints and mark the trapezoid to show that the altitude is perpendicular to the base.

2. Trapezoid $ADFL$ is shown with midsegment $\overline{WS}$ and altitude $\overline{TY}$.



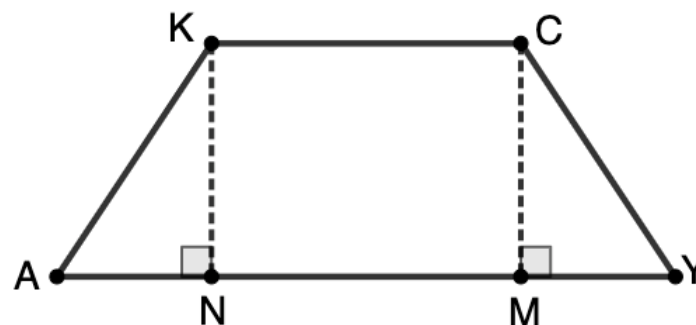
Select all of the information that is true about $ADFL$.

- | | | |
|--|--|--|
| ☐ $\overline{DF} \cong \overline{AL}$ | ☐ $m\angle LAD = m\angle FDA$ | ☐ $\overline{TY} \perp \overline{WS}$ |
| ☐ $\overline{AD} \parallel \overline{FL}$ | ☐ $\overline{WS} \parallel \overline{FL}$ | ☐ $m\angle AWS = m\angle DSF$ |
| ☐ $\overline{TY} \perp \overline{AD}$ | ☐ $\overline{DS} \cong \overline{SF}$ | ☐ $m\angle SFL = m\angle DSW$ |



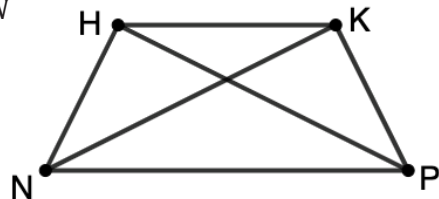
8.5.4 Guided Instruction: Proving Isosceles Trapezoid Properties

1. Trapezoid $KCYA$ is shown with altitudes $\overline{KN}$ and $\overline{CM}$.



- Trace the trapezoid onto a piece of patty paper.
- Fold the patty paper so that vertices K and C coincide.
- Describe what else you notice when you fold the patty paper.
- Work with a partner to list all of the congruent parts of trapezoid $KCYA$.
- Compare your list with that of another pair. Add any congruent parts you may have missed.
- Describe how the reflecting across the midpoint of $\overline{KC}$ shows that $KCYA$ can be classified as an isosceles trapezoid.

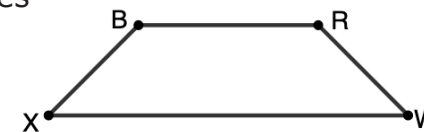
2. Given: $HKPN$ is an isosceles trapezoid with $\overline{HK} \parallel \overline{PN}$
 Prove: $\overline{HP} \cong \overline{KN}$



Complete the following two-column proof.

Statement	Reason
1.	1.
2. $\overline{HN} \cong \overline{KP}$	2.
3.	3. Base angles of an isosceles trapezoid are congruent
4. $\overline{NP} \cong \overline{NP}$	4.
5. $\triangle HNP \cong \triangle KPN$	5. SAS Congruence Theorem
6.	6. CPCTC

3. Given: $BRWX$ is an isosceles trapezoid with $\overline{BR} \parallel \overline{XW}$
 Prove: $\angle RBX$ and $\angle RWX$ are supplementary, $\angle BRW$ and $\angle BXW$ are supplementary



Given that $BRWX$ is an isosceles trapezoid, then

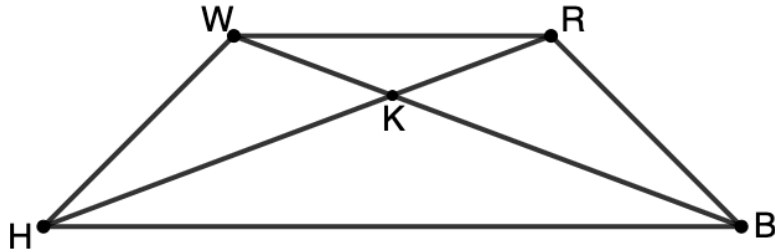
_____ because base angles of an isosceles trapezoid are congruent. Since it is given that _____, then $\angle RBX$ and $\angle BXW$ are _____ and _____ and _____ are supplementary because when parallel lines are cut by a transversal, _____ are supplementary. Therefore, by definition of supplementary angles, $m\angle BRW + m\angle RWX = \underline{\hspace{2cm}}$ and _____ = 180° . By substitution, $m\angle RBX + m\angle RWX = 180^\circ$ and _____.

Therefore, by _____, $\angle RBX$ and $\angle RWX$ are supplementary and $\angle BRW$ and $\angle BXW$ are supplementary.



8.5.5 Guided Practice: Isosceles Trapezoids

1. $WRBH$ is an isosceles trapezoid.

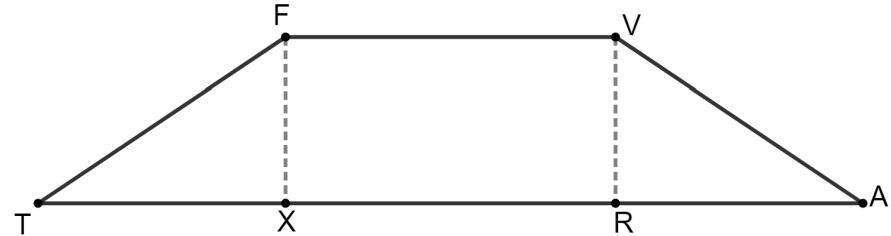


- If diagonal $\overline{WB} = 18$ and diagonal $\overline{RH} = 3x - 7$, find the value of x .
 - Diagonals $\overline{WB}$ and $\overline{RH}$ intersect at point K . If $m\angle RWB = 20.6^\circ$, $m\angle RHW = 24.4^\circ$, and $m\angle BKH = 138.9^\circ$, what is $m\angle WRB$?
2. Isosceles trapezoid $KCYA$ has altitudes $\overline{KN}$ and $\overline{CM}$. If $KC = 6$, $AY = 30$, and $KA = 15$, what is the length of altitude $\overline{CM}$?



8.5.6 Wrap-Up: Ticket Out the Door

1. Isosceles trapezoid $FVAT$ is shown with altitudes $\overline{FX}$ and $\overline{VR}$.



If $FT = 12$, $TA = 36\sqrt{5}$, and $VR = 8$, what is the length of $\overline{FV}$?

Unit 8, Lesson 6: Solving Real-World Problems With Quadrilaterals



Benchmarks

MA.912.GR.1.4 Prove relationships and theorems about parallelograms. Solve mathematical and real-world problems involving postulates, relationships, and theorems of parallelograms.

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.



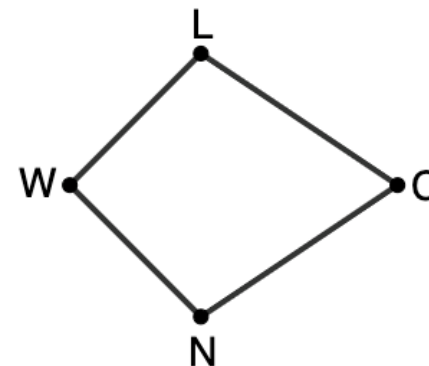
Learning Targets

- ☐ I can use properties of parallelograms to solve real-world problems.
- ☐ I can use properties of trapezoids to solve real-world problems.



8.6.1 Warm-Up: Error Analysis

Kite $WLCN$ is shown, with $m\angle W = 90^\circ$, $m\angle C = 68^\circ$, and $m\angle N = 6x + 5$.



Marcia and Eugene both worked to determine the value of x . Their work is shown.

Marcia	Eugene
$90 + 90 + 68 + 6x + 5 = 360$ $6x + 5 + 248 = 360$ $6x + 253 = 360$ $6x = 107$ $x = \frac{107}{6}$	$360 - (90 + 68) = 202$ $6x + 5 = 202$ $6x = 197$ $x = \frac{197}{6}$

- Evaluate each student's work, then write a note to each student explaining the error they made.

Marcia	Eugene



8.6.2 Activity: Word Problem Gallery Walk

1. Work through the word problems with your partner. Show your work and write your answers in the appropriate boxes below.

Problem Number	Work	Answer
1		$KR =$
2		Area =
3		$m\angle MGS =$ $m\angle PSG =$ $m\angle GMP =$ $m\angle MPS =$

4		$y =$
5		$FM =$ $RM =$ $TM =$
6		$DWFM$ _____ a parallelogram.

7		The dimensions of $KRWF$ are
8		$DW =$



8.6.3 Your Turn!

- I am a quadrilateral with two pairs of opposite parallel sides. At least one of my angles has a measure of 90° and both of my diagonals measure 12 inches. Select all shapes that match this description.
 - ☐ Parallelogram
 - ☐ Rectangle
 - ☐ Square
 - ☐ Rhombus
 - ☐ Kite
- In your own words, write a description of a parallelogram that could **only** match a square.
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rectangle?
- If a quadrilateral is a parallelogram, what additional information would be needed to show that the shape is a rhombus?



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

Properties of a parallelogram			
Properties of a rectangle			
Properties of a kite			
Completing proofs using properties and theorems about parallelograms			
Using the properties of a parallelogram to solve problems			
Properties of a trapezoid			
Using the properties of a trapezoid to solve problems			

Unit 8, Lesson 7: Trapezoid Midsegment Theorem



Benchmark

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.



Learning Targets

- ☐ I can prove the trapezoid midsegment theorem.
- ☐ I can use properties of trapezoids to solve mathematical and real-world problems.



8.7.1 Warm-Up: Trapezoids in Architecture

Four different buildings are shown.

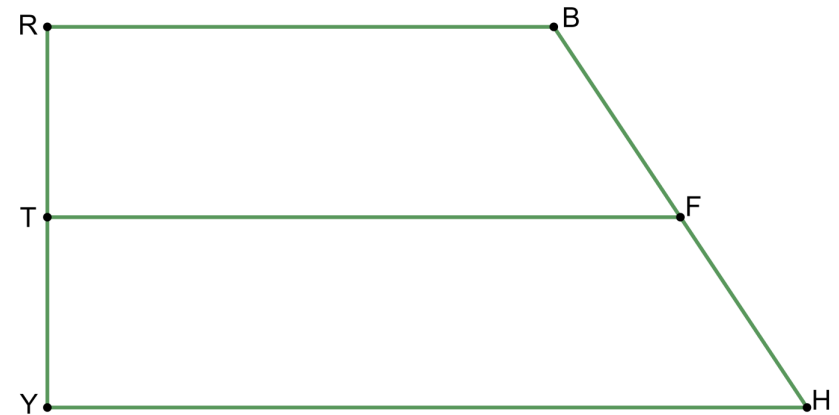
Machu Picchu, Peru	Greenland Center, Beijing, China
	
Garden Terrace, Nagasaki	International Spy Museum, Washington D.C.
	

1. What do you notice about the buildings?
2. Describe how a trapezoid is used in each building.



8.7.2 Exploration: Midsegment of a Trapezoid

1. Trapezoid $RBHY$ is shown, where $\overline{RB} \parallel \overline{YH}$.



- a. Use patty paper or a ruler to determine if $\overline{TF}$ is the midsegment of trapezoid $RBHY$. Justify your conclusion.
- b. Use a ruler to complete the table.

Segment(s)	Measure
$\overline{RB}$	
$\overline{YH}$	
$\overline{TF}$	

2. What do you notice about the relationship between the length of the midsegment and the sum of the lengths of the bases?

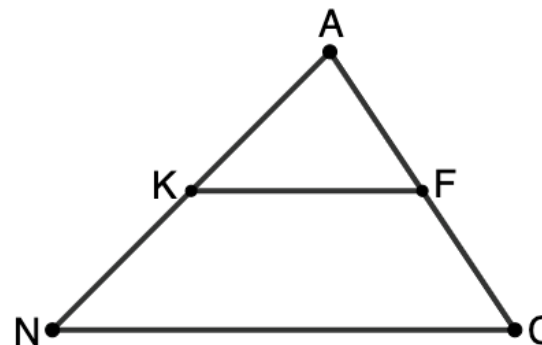
3. Write an equation to show the relationship between the length of the midsegment of a trapezoid and the length of its bases.



8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

The **midsegment** of a triangle is a segment joining the midpoints of two sides of a triangle.

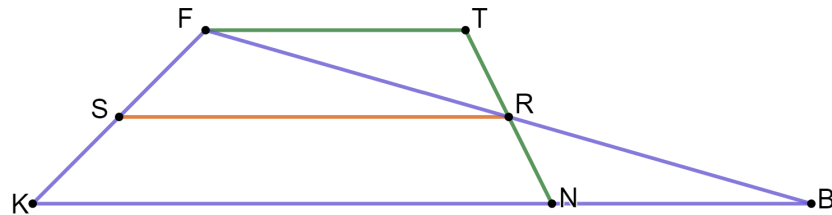
1. $\triangle ACN$ is shown with midsegment $\overline{KF}$.



Fill in the blanks to complete the triangle midsegment theorem.

The triangle midsegment theorem states that the midsegment of a triangle is _____ to the third side it does not touch and is _____ the length of the third side.

2. $\triangle FBK$ with midsegment $\overline{SR}$ and trapezoid $FTNK$ are shown where $\overline{FT} \parallel \overline{KN}$.



a. Explain why $SF = KS$.

b. If $\overline{NB} \cong \overline{FT}$, justify why $\overline{NR} \cong \overline{RT}$.

c. Discuss with your partner what information is necessary so that the definition of a trapezoid can be used to state that $FTRS$ is a trapezoid. Summarize your discussion.

- d. Discuss with your partner what else needs to be determined so that it can be stated that $\overline{SR}$ is a midsegment of $FTNK$. Summarize your discussion.



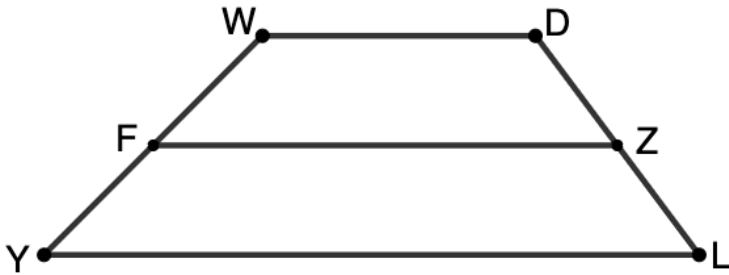
Trapezoid Midsegment Theorem

The midsegment of a trapezoid is parallel to the bases and is half the sum of the bases.



8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

1. Trapezoid $WDLY$ with midsegment $\overline{FZ}$ is shown.

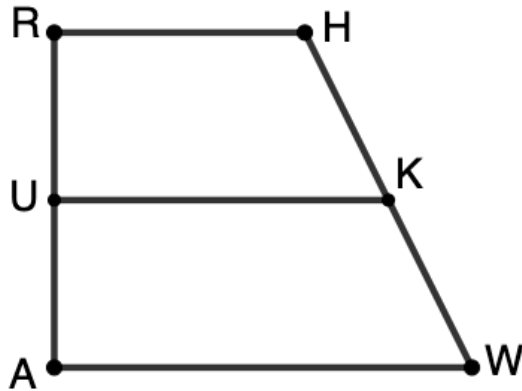


- a. If $WD = 14.2$ and $YL = 29.7$, what is the length of $\overline{FZ}$?
- b. If $WDLY$ is an isosceles trapezoid and $m\angle DLY = 53^\circ$, what is $m\angle WFZ$?
- c. If $FZ = 11.4$ and $YL = 17.6$, what is the length of $\overline{WD}$?
- d. What is the value of k if $WD = 4k + 4.2$, $FZ = 7k + 2.1$, and $YL = 6k + 16.8$?
- e. If $WF = \frac{2}{3}p$ and $WY = 12 - 2p$, what is the length of $\overline{WY}$?

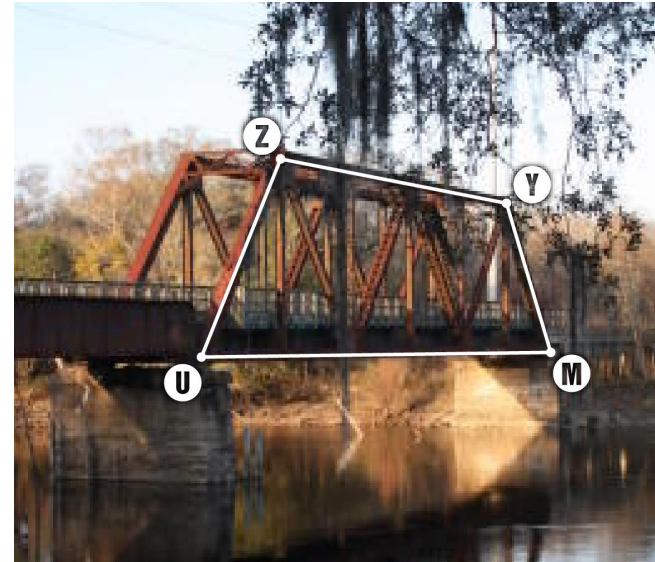


8.7.5 Your Turn!

1. Trapezoid $RHWA$ is shown with midsegment $\overline{UK}$. Find the values of x and y if $RU = -2x - 2y$, $AU = 10$, $RH = 4x$, $UK = 14$, and $AW = -y + 8$.



2. Located on the Nature Coast State Trail in north-central Florida is the Suwannee River Trestle, a bridge that is shaped like an isosceles trapezoid. The state has been inspecting older bridges to ensure their safety and has determined that the Suwannee River Trestle, represented as trapezoid $ZYMU$, needs an additional support brace through its middle.



The support brace is to connect sides $\overline{ZU}$ and $\overline{YM}$ at their midpoints. If the length of leg $ZU = 28$ feet (ft) and the perimeter of the trestle measures 153 ft, what is the length, in ft, of the new support brace?



8.7.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand the trapezoid midsegment theorem.

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